

Awesome Web Testing with **Playwright** and AI



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Who here...



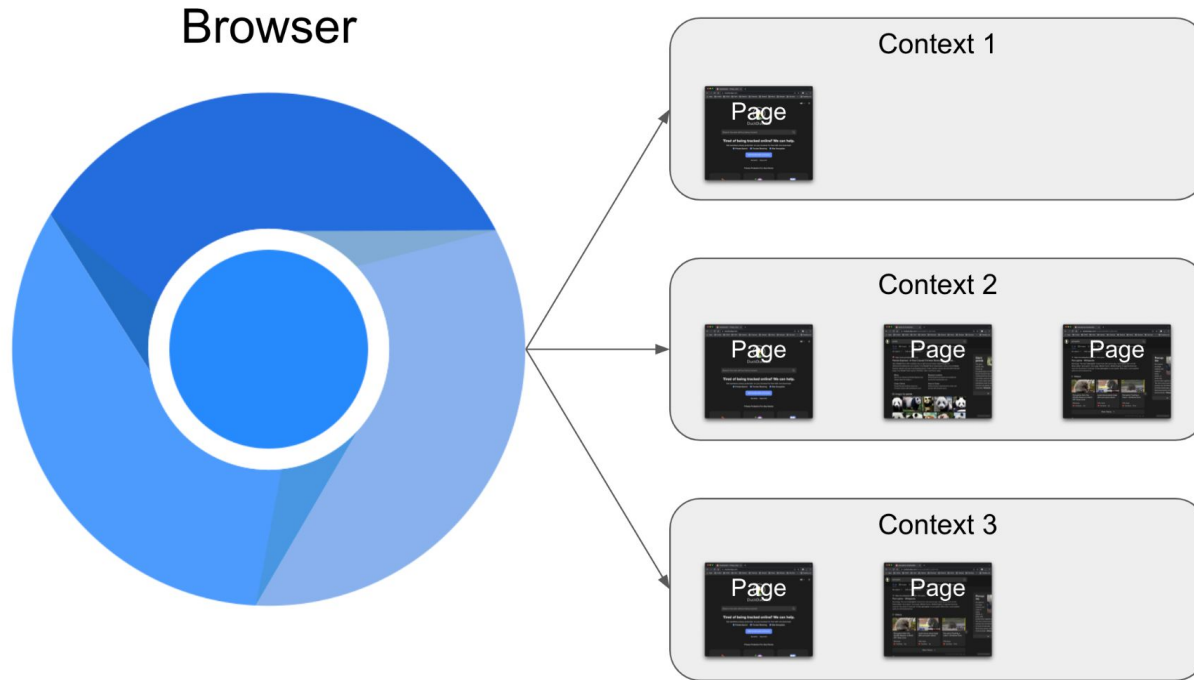
- Knows what Playwright is?
- Has automated tests with Playwright?
- Has used AI to write Playwright tests?
- Has used Playwright MCP with agents?
- Has used Playwright CLI with skills?

Playwright snapshot:

- A modern, open source web test framework from Microsoft
- Manipulates the browser via (superfast) debug protocols
- Works with Chromium/Chrome/Edge, Firefox, & WebKit
- Provides automatic waiting, test generation, UI mode, etc.
- Can test UIs and APIs together
- Bindings for Node (JS/TS), Python, Java, & .NET (C#)
- AI support includes MCP + agents and CLI + skills



Browsers, contexts, and pages



Browser automation tool comparison

Selenium 	Cypress 	Playwright 
Popular among testers	Popular among developers	So hot right now
C#, Java, JavaScript, Ruby, Python	JavaScript only	C#, Java, JavaScript, Python
All major full browsers	Chrome, Edge, Firefox, Electron	Chromium+, Firefox, WebKit
WebDriver protocol	In-browser JavaScript	Debug protocols
Just a tool – requires a framework	Full, modern framework	Full, modern framework
Can be slow/flaky if waiting is poor	Visual execution but can be slow	Typically the fastest execution
Open source, standards, & gov	Open source from Cypress	Open source from Microsoft

The Building Blocks of **AI-powered Development**

Coding Agent: Orchestrates a model, rules, sub-agents, and tools to produce code end-to-end

Model: The LLM responsible for all reasoning, planning, and generation

Rule: A standing instruction in the system prompt that governs agent behavior unconditionally at all times

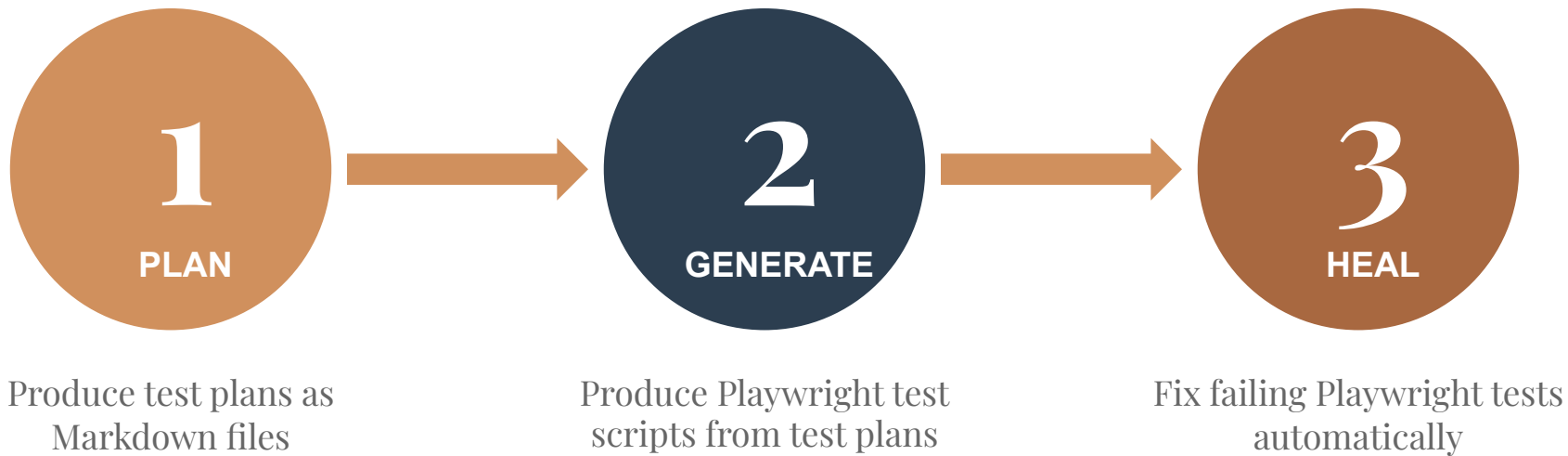
Sub-agent: A spawned worker with its own context window, tools, directives, and goal to enable isolated work

Skill: Packaged prompt and instructions giving an agent a specific, well-defined capability within a context window

MCP: Model Context Protocol — open standard for connecting agents to external servers, APIs, and data sources

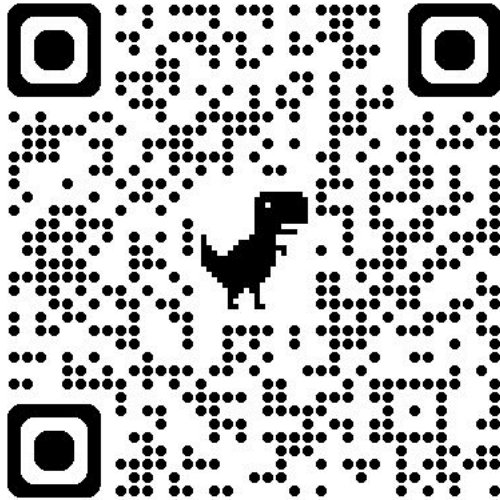
CLI: Local command-line tools the agent invokes to run code, manage files, and interact with the system

The AI Testing Process



Mirrors BDD: Discovery → Formulation → Automation

Live Code Repository



[https://github.com/
AutomationPanda/
buggyboard-web-app](https://github.com/AutomationPanda/buggyboard-web-app)

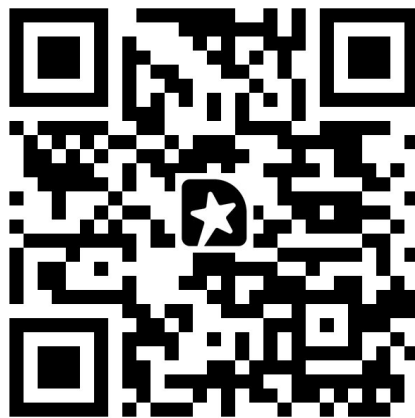
Live coding results will be pushed
to the **qaoth** branch.

Playwright with AI



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Feedback Form



Awesome Web Testing with Playwright and AI

<https://github.com/AutomationPanda/buggyboard-web-app>